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Performance of peripheral catheters inserted with ultrasound guidance versus landmark technique after a simulation-based mastery learning intervention

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Abstract

Problem: Ultrasound-guided peripheral intravenous catheter (USGPIV) insertion is an effective method to gain vascular access in patients with difficult intravenous access (DIVA). While USGPIV success rates are reported to be high, some studies have reported a concerning incidence of USGPIV premature failures.

Aims: The purpose of this study was to compare differences in USGPIV and landmark peripheral intravenous catheter (PIV) utilization and failure following a hospital-wide USGPIV training program for nurses.

Methods: The authors performed a retrospective, electronic medical record review of all USGPIVs and PIVs inserted at a tertiary, urban, academic medical center from September 1, 2018, through September 30, 2019. The primary outcome was differences between USGPIV and PIV time to failure.

Results: A total of 43,470 short peripheral intravenous catheters (PIVCs) were inserted in 23,713 patients. Of these, 7972 (16.8%) were USGPIV. At 30 days of follow-up, for PIVCs with an indication for removal documented, USGPIVs had higher Kaplan–Meier survival probabilities than PIVs ($p < 0.001$).

Conclusions: The use of simulation-based mastery associated with USGPIVs, demonstrated lower failure rates than standard PIVs after 2 days and USGPIVs exhibited improved survival rates in patients with DIVA. These findings suggest that rigorous simulation-based insertion training demonstrates improved USGPIV survival when compared to traditional PIVCs. SBML is an extremely useful tool to ensure appropriately trained clinicians acquire the necessary knowledge and skillset to improve USGPIV outcomes.

Keywords

Ultrasound, venous access, simulation, difficult intravenous access, device failure, outcomes

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Introduction

Peripheral intravenous catheters are the most common form of intravenous (IV) access worldwide and a cornerstone of inpatient care. Short peripheral IV catheters (PIVC) are catheters measuring less than 7.5 cm in length.[1] An estimated 59%–86% of hospitalized patients receive at least one PIVC during their admission.[1–4] Despite their ubiquity, PIVC insertion may be challenging or ultimately unsuccessful in some patients, a condition known as difficult intravenous access (DIVA).[5–8] An estimated 8%–59% of hospitalized patients are affected by DIVA.[6,8–12] Patients with DIVA are vulnerable to delays in care, undergo multiple percutaneous punctures, experience higher pain scores, and may receive more invasive vascular access devices including central venous access devices (CVAD)s.[6,8,13–20]

Over the last two decades, ultrasound-guided peripheral intravenous catheter (USGPIV) insertion has emerged as an effective method to establish venous access in patients with DIVA. USGPIV insertions utilize point-of-care ultrasound (POCUS) to dynamically guide PIVCs.[21] Compared to traditional methods of PIVC insertion, USGPIV has higher success rates, improved patient satisfaction rates, and yields faster time to access.[15,18,22–25] USGPIV training programs have also been shown to reduce unnecessary CVAD use.[16,17,26,27]

Despite the known efficacy of USGPIV to establish access in patients with DIVA, several studies have raised concerns about the longevity of USGPIVs. These studies found that USGPIVs have a high incidence of premature failure when compared to traditional (anatomic landmark inserted) PIVCs.[28–31] Reports of USGPIV performance vary, but range from an 8% failure rate within the first hour after insertion to 47% within 24-h[29,31] Other studies report only 19% of USGPIVs to survive past 72 h with a median survival of 1.25 days and a mean overall dwell time of 1.2 days.[29,30,32] PIVC failures can lead to increased costs, delays in care, complications from device failure, such as infiltration, phlebitis, or thrombosis, and often expose the patient to additional pain and trauma from recurrent PIVC insertion attempts.[33–37] There are also increasing concerns that repeated puncture attempts and recurring device failure precipitate venous depletion, and undermines overall vessel health.[1,38] These reports have led some authors to conclude that USGPIVs may only be appropriate for short-term infusions or as a temporary bridge until more definitive vascular access can be established.[29–31]

The purpose of this study was to compare the survival rate of USGPIVs to traditional PIVCs, following dissemination of a hospital-wide USGPIV simulation-based mastery learning (SBML) curriculum for nurses.[39] Simulation-based mastery learning is an extreme form of competency-based learning where all participants are required to meet or exceed a high standard on a simulated skills assessment before completion of training. Those who do not meet this standard participate in further practice until they can be retested and meet the standard. We hypothesized that after the USGPIV SBML training initiative, USGPIVs would demonstrate equivalent survival rates compared to traditional PIVCs and improved survival compared to prior published reports.

Methods

Study design and setting

This was a single-center, retrospective, observational study that took place at Northwestern Memorial Hospital (NMH), a large, urban, academic medical center. Data was captured from the electronic medical record (EMR) for all documented PIVCs inserted dwelling up to 30 days from September 1, 2018, through September 30, 2019. In the year preceding this study (April 2017–August 2018), a hospital-wide USGPIV SBML curriculum had been established for bedside nurses.[39] This study was approved by the Northwestern University Institutional Review Board.

PIVC insertion procedures

At NMH, nurses are responsible for the insertion and maintenance of PIVCs. When a patient requires a new PIVC, the nurse first attempts a traditional PIVC insertion using a combination of inspection, palpation, and anatomic landmarks (hereafter referred to as a PIV insertion). If no suitable vessels are identified or if the patient had undergone two or more PIV insertion attempts, the patient meets the criteria for DIVA, and becomes eligible for USGPIV insertion.[6,8–11] USGPIV insertion is then requested from a nurse who had completed the SBML training program and satisfied pre-established competencies. In the event of ongoing insertion failures, additional resources to establish vascular access could be sought from the facility's vascular access team or physicians.

Nurses are responsible for the documentation of all PIVCs. Standard NMH documentation practices include date and time of insertion, clinical indication, anatomical location of the inserted device, the use of ultrasound guidance, identification of the inserter, date and time of removal, and indication for removal (IFR). According to institutional policy, PIVC sites are assessed and documented by nursing staff every 8 h. PIVCs are removed when clinically indicated, that is, when there is evidence of device occlusion, pain or tenderness, redness, infiltration, swelling, leakage, phlebitis, edema, induration, color, or temperature change or if requested by patients or family members.[24] PIVCs are not routinely exchanged at a given time interval without a clinical indication. At the time of this study, the institutional practice was such that all patients routinely had a vascular access device (VAD) present throughout the duration of their hospitalization, unless

a treating physician authorized the patient to be without IV access.

SBML training

Institutional training for USGPIV insertion was provided through voluntary enrollment in an SBML curriculum described in detail elsewhere.[39] Briefly, this curriculum occurred over 3 phases. Phase 1 consisted of a 4-h session featuring a simulated skills pretest followed by both didactic instruction and a hands-on skills lab. During the didactic session, nurses were trained on hygienic practices, appropriate target vessel selection (depth, diameter, anatomic location), appropriate catheter selection, documentation practices, and critical insertion techniques. During the skills lab, participants deliberately practiced these USGPIV insertion skills on a simulator. In phase 2 nurses returned to complete a simulated skills posttest and were required to meet or exceed a predetermined minimum passing standard (MPS). Nurses who did not meet the MPS were given feedback and additional opportunities to practice until they were able to meet this standard at retesting. Finally, in phase 3, nurses were required to complete 5 successful insertions on real patients with DIVA while supervised by a trained proctor. A cohort of nurses from each of NMH's inpatient units completed all phases of the curriculum and were certified to independently perform USGPIV insertions. At the beginning of the study period 160 nurses completed the SBML USGPIV training whereas, by the end, this number grew to 220 nurses.

Data collection

Data was collected using the Northwestern University Enterprise Data Warehouse (EDW). The EDW is a database of clinical information extracted from the electronic medical records at NMH that can be queried using Structured Query Language syntax.[40] We queried the EDW for patients with PIVCs who were ⩾18 years old, had their PIVC inserted while in the Emergency Department (ED), observation unit, or inpatient units, and were admitted to the hospital with inpatient status. PIVCs inserted in obstetrical, neonatal, interventional, surgical, and outpatient departments were excluded, as were PIVCs inserted in patients who were ultimately registered as outpatient or observation status. The EDW query included information about the date and time of each patients' admission, PIVC insertion, PIVC removal and discharge, use of ultrasound guidance during PIVC insertion, and the patient's age, sex, body mass index (BMI), hospital unit at the time of insertion, anatomical location of the PIVC device, and PIVC gauge.

We analyzed the data in two ways. First, survival analyses were done to evaluate PIVC failures between USGPIV to PIVs. For this analysis, we excluded all PIVCs with missing data regarding the IFR. PIVCs were categorized as either a "failure" or "acceptable removal" according to the IFR. PIVCs removed for reasons including leakage/drainage, occlusion, infiltration, pain, phlebitis, patient/family request, removed by patient and other) were categorized as failures. PIVCs with an IFR of "treatment complete" were considered acceptable removals. Second, to account for the potential differences in PIVCs with missing IRF data, frequencies for all-cause removal of PIVCs were compared between the PIV and USGPIV cohorts at a variety of time points (24, 48, 72, 96, and 120 h) for patients with a LOS greater than each time cutoff.

Statistical analysis

PIVC survival time was used to estimate a Kaplan-Meier (KM) survival probabilities at 24, 48, 72, 96, and 120 h after insertion. A log rank test was used to determine the significance of differences between PIV and USGPIV in overall time to failure at 30 days. A 30-day cut-off was used because it had been anecdotally reported at this institution that, occasionally, PIVCs had survived up to 1 month. A Cox proportional hazards model was used to estimate hazards ratios and 95% confidence intervals between failure times of USGPIV and PIV insertions, adjusted for clustering of multiple PIVCs by the same patient as well as patient's age, sex, body mass index, length of stay (LOS), hospital unit at the time of insertion, anatomical location of the PIVC device, and PIVC gauge. Chi square tests were used to determine the significance of differences between PIV and USGPIV for all-cause removal at 24, 48, 72, 96, and 120 h after insertion. Statistical analysis was performed with IBM SPSS Version 25 (Armonk, NY) and Stata Version 15 (College Station, TX) for all data analyses.

Results

A total of 43,470 PIVCs inserted in 23,713 patients were included in the study. Of these PIVCs, 7972 (16.8%) were USGPIVs. The characteristics of patients and PIVCs are summarized in Table 1. Patients with higher LOS tended to have higher proportions of USGPIVs. USGPIVs were used more frequently in the intensive care units (ICUs), constituting 32.8% of all PIVCs inserted compared with 14.5%, 15.0%, and 15.3% utilization in oncology, medical/surgical units, and ED settings respectively. The majority of USGPIVs were inserted in the antecubital fossa or more distally (73.5%), whereas the forearm was utilized in 44.0% of USGPIV insertions.

The KM survival analysis was performed on 5480 (68.7%) of USGPIVs and 27271 (69.0%) of PIVs where the IFR was documented. Survival probabilities of USGPIV and PIV are summarized at various time points in Figure 1 and Table 2. USGPIVs had slightly lower survival probabilities than PIVs within the first 24 h (0.82 vs 0.86)

Table 1. Demographic and clinical characteristics of ultrasound-guided peripheral intravenous catheter (USGPIV) and traditional peripheral intravenous catheter (PIV) patient insertions.

Category	USGPIV insertions $n = 7972$ (%)	PIV insertions $n = 39{,}498$ (%)
Age		
18–35	1213 (15.2)	9106 (23.1)
36–55	2100 (26.3)	9568 (24.2)
56–64	1473 (18.5)	5747 (14.6)
65–84	2756 (34.6)	12,292 (31.1)
⩾85	430 (5.4)	2785 (7.1)
Gender (Male)	3423 (42.9)	16,184 (41.0)
BMI		
Missing	379 (4.8)	2172 (5.5)
< 18.5	360 (4.5)	1663 (4.2)
18.5–24.9	2143 (26.9)	10,967 (27.8)
25–29.9	2006 (25.2)	11,973 (30.3)
30–39.9	2090 (26.2)	9906 (25.1)
⩾40	994 (12.5)	2817 (7.1)
Hospital LOS		
1–2 days	294 (3.7)	3375 (8.5)
3–4 days	1812 (22.7)	16,856 (42.7)
5–7 days	1418 (18.6)	6525 (16.5)
8–14 days	2007 (25.2)	6719 (17.0)
⩾15 days	2378 (29.8)	6023 (15.2)
Hospital unit		
Oncology	306 (3.8)	1807 (4.6)
ICU	1528 (19.2)	3126 (7.9)
ED	1986 (24.9)	11,025 (27.9)
Medical/surgical	4152 (52.1)	23,540 (59.6)
Catheter anatomic location		
Forearm	3506 (44.0)	16,573 (42.0)
Upper Arm	2010 (25.2)	1113 (2.8)
Antecubital	2193 (27.5)	11,500 (29.1)
Hand	71 (0.9)	5581 (14.1)
Wrist	85 (1.1)	4014 (10.2)
Other*	107 (1.3)	717 (1.8)
Catheter gauge (Ga.)		
14–16	145 (1.8)	3408 (8.6)
18	2180 (27.3)	21,019 (25.8)
20	5432 (68.1)	19975 (50.6)
22–24	84 (1.1)	5008 (12.7)
Missing	131 (1.6)	915 (2.3)
IFR documented	5480 (68.7)	27271 (69.0)

BMI: body mass index; LOS: length of stay; IFR: indication for removal.
($N = 47470$ for 23,713 patients).
*External Jugular, Leg, or Foot.

and the same survival probability at 48 h (0.70 vs 0.70). However, at each time period after 48 h, USGPIVs had higher survival probabilities than PIVs with the overall curve at 30 days showing significantly increased survival of USGPIVs compared to PIV ($p < 0.001$).

The Cox regression analysis confirmed that USGPIVs were less likely to fail compared to PIVs (HR 0.91 (95% CI 0.87–0.95)) after controlling for multiple clinical characteristics (Table 3). The forearm USGPIV insertion site was less likely to fail compared to all other sites (HR 0.78 (95% CI 0.73–0.82)).

The all-cause removal frequencies for PIVs and USGPIVs, with and without IFR documented, are summarized in Table 4. At 24 h, 80% of USGPIVs and 81.9% of PIVs remained inserted for patients with LOS >24 h ($p < 0.001$). However, at 48, 72, 96, and 120 h, a greater percentage of USGPIVs remained inserted compared with traditional PIVs for patients that had a LOS exceeding

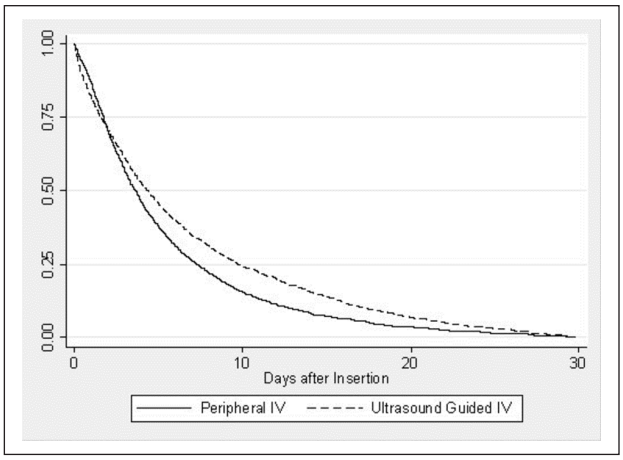

Figure 1. Kaplan–Meier survival probability comparing ultrasound-guided peripheral intravenous catheter (USGPIV) and traditional peripheral intravenous catheter (PIV) patient insertions up to 30 days after insertion. USGPIVs lasted longer than PIV (log-rank test; $p < 0.001$).

Table 2. Kaplan–Meier survival probabilities of ultrasound-guided peripheral intravenous catheter (USGPIV) and traditional peripheral intravenous catheter (PIV) patient insertions at different follow-up times for catheters with documented indications for removal.

Time cut-off	USGPIV $n = 5480$	PIV $n = 27{,}271$
24 h	0.82 (0.80–0.82)	0.86 (0.86–0.87)
48 h	0.70 (0.69–0.72)	0.70 (0.69–0.71)
72 h	0.61 (0.60–0.62)	0.57 (0.56–0.57)
96 h	0.52 (0.51–0.54)	0.46 (0.45–0.46)
120 h	0.47 (0.44–0.47)	0.38 (0.37–0.38)

[Probability, (95% Confidence Interval)]. $N = 32{,}751$. $p < 0.001$ for entire Kaplan–Meier curve up to 30 days.

each time point cut off ($p < 0.001$). The results of this analysis were similar to the findings of the KM analysis suggesting that outcomes were similar whether or not the IFR was documented.

Discussion

To our knowledge, this research report describes the largest observational study of PIVC performance to date and represents the first study evaluating USGPIV performance after initiating a hospital-wide training initiative for bedside nurses. This study has two novel findings that advance what is known about the longevity of USGPIVs. First, the study showed that USGPIVs were less likely to fail than traditional PIVs. This contradicts a prior published report showing a higher incidence of USGPIV premature failures when compared to standard PIVs.[30] Secondly, this study also found that overall USGPIV survival rates were superior than previously reported. In other previous studies, 24 h USGPIV survival rates of between 53% and 76% had been reported, while this study found a 24 h KM survival rate of 83%.[28,29] Similarly, previous studies reported a 48 h survival rate of 63%, and a 19% survival rate at 72 h, while this study found USGPIV survival rates of 70% at 48 h, and 61% at 72 h.[28,29] Considered collectively, the findings challenge previous research that suggests USGPIVs perform poorly and should only be used as a rescue procedure or as a bridge to more definitive vascular access.[29–31]

Several factors have been hypothesized to impact USGPIV survival including catheter length, the amount of catheter residing within the vein following insertion, vessel depth and diameter, and anatomic location.[28,32,41,42] With regard to catheter length, several studies have explored the approach of using longer catheters to improve catheter survival. While this approach does appear to improve device longevity, there have been concerns that longer catheters may increase risk of phlebitis and infections as well as incur higher healthcare costs.[43–46] In addition to the effect of catheter length, the amount of catheter residing within the vein can be optimized to improve survival by using an appropriate insertion angle that takes vessel depth and diameter into account.[28,41] Finally, the anatomic location of the vein used for USGPIV insertion likely affects catheter survival.[28] Anatomic locations with more abundant and mobile subcutaneous tissues, deeper vessels, and areas of flexion (antecubital fossa) or that come in contact with the chest wall (medial upper arm) are more prone to catheter failure.[1,28,36] Therefore, forearm vessels are considered more appropriate vessels for USGPIV insertions as demonstrated by the better survival rates reported in this study. Furthermore, the use of distal vessels helps to preserve more proximal vessels for instances when more long-term catheters such as CVADs are needed.[1]

Our USGPIV SBML curriculum contained rigorous training about catheter length, the amount of catheter residing within the vein following insertion, vessel depth and diameter, catheter to vein ratio and anatomic location. During both the didactic and hands-on practice sessions nurses were educated on the length of catheter residing within the vein by using the longer available 4.78 cm catheter at a 45° angle of insertion. Vessel selection criteria were emphasized to maximize the use of both large and shallow vessels ($\geqslant$0.3 cm diameter $\leqslant$0.5 cm deep) and to prioritize distal vessels (forearm), avoiding the deep brachial and basilic vessels whenever clinically possible. Knowledge and USGPIV insertion skills were rigorously tested using an objective scoring system (skills checklist) until a very high level of skill was achieved during both Phases 2 (simulated insertions) and 3 (patient insertions) of the curricula.

While there has been increased attention on how devices and vessel characteristics affects USGPIV survival, little

Table 3. Cox analysis showing hazard ratios and 95% confidence intervals for ultrasound-guided peripheral intravenous catheter (USGPIVs) compared to the traditional peripheral intravenous catheter (reference) adjusted for demographic and clinical characteristics in catheters with documented indications for removal.

Risk factor	Hazard ratio, (95% confidence interval). $n = 32{,}713$
USGPIV	0.91 (0.87–0.95)
Age	
18–35	Reference
36–55	1.02 (0.97–1.08)
56–64	0.97 (0.91–1.03)
65–84	1.02 (0.96–1.08)
⩾85	1.21 (1.11–1.31)
Gender (Male)	0.92 (0.89–0.96)
BMI	
<18.5	0.98 (0.91-1.06)
18.5–24.9	Reference
25–29.9	0.92 (0.88–0.96)
30–39.9	0.95 (0.91–1.0)
⩾40	1.0 (0.93–1.06)
LOS	
1–2 days	Reference
3–4 days	0.82 (0.74–0.90)
5–7 days	0.81 (0.74–0.90)
8–14 days	0.71 (0.64–0.79)
⩾15 days	0.60 (0.54–0.67)
Hospital unit	
Oncology	0.84 (0.78-0.91)
ICU	0.81 (0.76-0.86)
ED	Reference
Medical/surgical	0.90 (0.86-0.94)
Catheter anatomic location	
Forearm	0.78 (0.73–0.82)
Upper arm	Reference
Antecubital	0.88 (0.82–0.93)
Hand	1.0 (0.93–1.07)
Wrist	0.91 (0.84–0.97)
Other*	1.63 (1.37–1.93)
Catheter gauge (Ga.)	
14–16	0.78 (0.72–0.85)
18	0.92 (0.89–0.96)
20	Reference
22–24	1.21 (1.16–1.28)
Missing	1.12 (1.01–1.24)

BMI: body mass index; LOS: length of stay.
*External Jugular, Leg, or Foot.

Table 4. Percent of all ultrasound-guided peripheral intravenous catheter (USGPIV) and traditional peripheral intravenous catheter (PIV) patient insertions ($N = 47{,}470$) remaining in place at different follow-up times.

Time cutoff	USGPIV n (%) $n = 7972$	PIV n (%) $n = 39{,}498$
24 h*	6371 (80.0)	32,340 (81.9)
48 h*	5050 (63.8)	20,930 (53.5)
72 h*	3956 (51.7)	13,501 (37.7)
96 h*	3075 (43.6)	9190 (32.7)
120 h*	2395 (37.4)	6468 (28.3)

Each comparison excludes lengths of stay that were less than dwell time cut-offs.
*$p < 0.001$.

has been published about how training on technical skills affect USGPIV catheter longevity.[47] It is hypothesized that poor insertion technique can cause vessel trauma, which can, in turn, lead to localized hematoma, pain, leaking, infiltration, occlusion, infection, and traumatic thrombophlebitis.[1,48–50] The USGPIV SBML curriculum rigorously trained and assessed nurses' technical skills including needle tip control and localization, maintaining the needle tip within the center of the lumen (prevention of transfixing the vessel and inadvertent nerve/arterial injury), and confirmation of the catheter location using ultrasound visualization. As SBML has been shown to improve skills and patient outcomes for a variety of procedures in medical trainees including ultrasound-guided CVAD insertion, it is not surprising that our curriculum also improved outcomes for USGPIV insertions.[51–57] Incorporation of evidence-based best practices into the SBML curriculum may have contributed to the improved catheter survival seen in actual patient care. We believe SBML should be used to rigorously train nurses to a high standard of USGPIV skills, and replication of our findings should be an area for future inquiry.

Limitations

This study has several limitations. First, this is a single-center, retrospective observational study potentially limiting its generalizability across other facilities. Second, it is possible that a small number of USGPIVs and PIVs may have been inserted by clinicians other than the trained nurses who completed SBML competency. Third, due to some omissions in the EMR documentation, a percentage of PIVCs from both the USGPIV and PIV cohort did not have an IFR and therefore could not be included in the KM analysis. However, when all-cause PIVC removal were explored for the entire cohort (regardless of IFR), the findings were similar to the KM analysis. Finally, other important vascular access device complications, such as catheter-related infection (CRI) and catheter-related venous thrombosis (CRVT) were not evaluated and require further investigation.

Conclusions

USGPIVs exhibited a better survival rate (i.e. lesser device failure) inserted in patients with DIVA when compared to traditionally inserted, standard PIVs. While USGPIVs had

slightly lower survival than standard PIVs at 24 h, these devices demonstrated significantly improved survival rates when compared to a standard PIV out to 30 days. This may have been due to device insertions in the antecubital fossa region and subject to mechanical failures from flexion. Forearm vessels were utilized in only 44% of USGPIV insertions and despite clearly low failure rates, there is still a predominance for clinicians to utilize sites that are not optimal. The forearm USGPIV insertion site was reported to be less likely to fail when compared to all other studied sites. Our findings suggest that rigorous simulation-based insertion training demonstrates improved USGPIV survival compared to traditional PIVCs. SBML is an extremely useful tool to ensure appropriately trained clinicians acquire the necessary knowledge and skillset to improve USGPIV outcomes.

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